

# Findings-to-Fix

Fix confirmed Checkmarx findings from your terminal or editor with Claude Code

Built on Checkmarx One Triage Assist and Remediation Assist. Triage Assist has already evaluated the findings on the platform, using reachability, exploitability, code context, and policy signals, and confirmed the ones that require action. Findings-to-Fix takes only those, asks Remediation Assist to generate the review-ready fix, and shows you each change in Claude Code as a diff to accept or reject. The agent proposes; you approve.

## What you need

- **A Checkmarx One API key** on your machine, in your shell profile or in the `cx` CLI configuration.
- **Python 3 or Node** on your machine. Either one works.
- **Claude Code** in a terminal, or through its VS Code or JetBrains extension.

## Contents

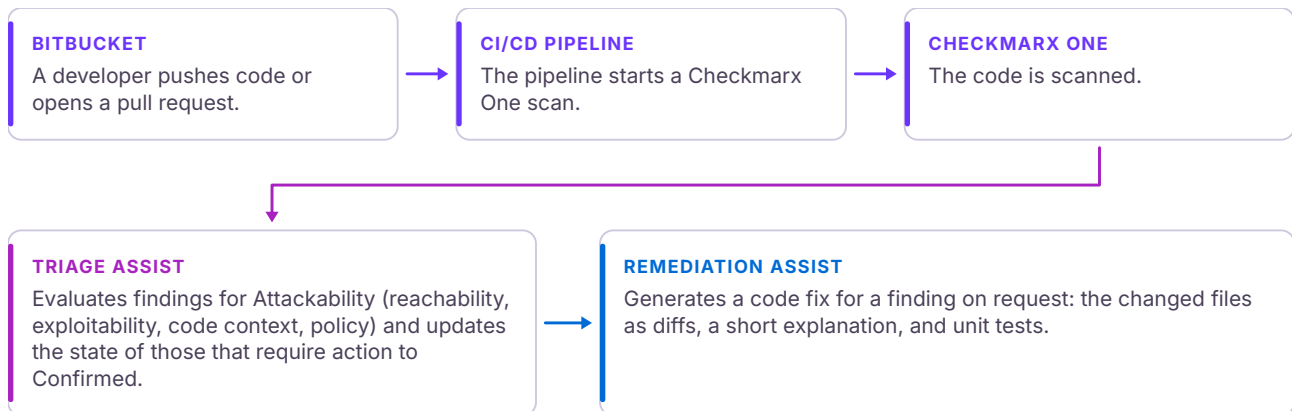
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Only findings that Checkmarx Triage Assist has marked **Confirmed**, at critical or high severity, are fixed. Nothing is committed or pushed. You review, then commit as usual.

# How it fits together

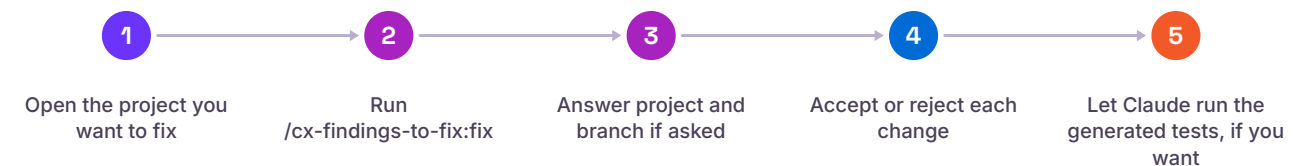
Three views of the same flow: what the platform does after a scan, what a developer does at the keyboard, and the path a single fix takes.

## 1 After a scan



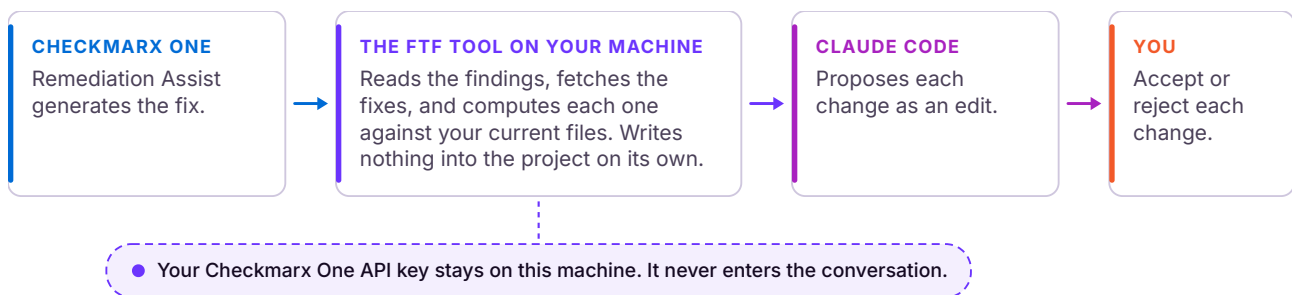
All of this happens on the Checkmarx One platform. No developer action is needed for any of it.

## 2 What the developer does



Five steps in Claude Code. Type `/cx-f` and press Tab to complete the command name. Adding the project and branch after it skips step 3.

## 3 How a fix travels



Every change to code arrives as an edit that you accept or reject. The plugin never commits or pushes.

# Getting started

Three things once. After that it is one command.

## 1 A Checkmarx One API key on your machine

Ask your Checkmarx admin for an API key. Then either put it in your shell profile:

```
export CX_APIKEY=<your key>
```

or, if you already use the Checkmarx `cx` command line tool:

```
cx configure set --prop-name cx_apikey --prop-value <your key>
```

Findings-to-Fix reads it from either place. It never shows the key to Claude.

## 2 Python or Node on your machine

Either one works. Most Macs and Linux machines already have Python 3. On Windows, if you have `python3` or `node` in a terminal, you are set.

## 3 Install the plugin

```
claude plugin marketplace add cx-israel-ogunsakin/cx-findings-to-fix-claude  
claude plugin install cx-findings-to-fix@cx-findings-to-fix-claude
```

For a private copy of the repository, use its Git URL or a local folder path in the first command. Claude Code's Bash tool has to be allowed, and the first run asks you to permit the plugin's command.



```
claude - ProjectHub9  
  
$ claude  
  
Findings-to-Fix is installed. Run /cx-findings-to-fix:fix in  
the project you want to fix (type /cx-f and press Tab).  
  
> /cx-findings-to-fix:fix CxRW-Sandbox/ProjectHub9 feat/update-routes
```

Type `/cx-f`, press Tab, and add the project and branch if you know them

# Using it

Open a project, run one command, then review each change.

## 1 Open the project you want to fix and start Claude Code

Run `claude` in a terminal in the project folder. The Claude Code extensions for VS Code and JetBrains work the same way.

## 2 Type `/cx-f`, press Tab to complete, then Enter

Add the project name and branch after the command to skip the questions: `/cx-findings-to-fix:fix CxRW-Sandbox/ProjectHub9 feat/update-routes`

### Claude will

- work out which Checkmarx project and scan this is. If the project name does not match your git setup, it lists the likely projects and asks you to pick. If only one branch has scans (common for monorepos and zip uploads, which Checkmarx files under the branch name `.unknown`), it uses that and tells you. If several branches have scans and yours is not one of them, it lists the most recently scanned ones with dates and asks. It never guesses.
- fetch the fixes from Checkmarx: a couple of minutes the first time, seconds after that.
- show you a table of the confirmed findings and ask which to apply.
- propose each change as an edit you accept or reject, file by file.
- explain what each fix does and why, then offer to run the tests Checkmarx generated.

Nothing is committed or pushed. You review, then commit as usual.

```
claude - ProjectHub9

> /cx-findings-to-fix:fix CxRW-Sandbox/ProjectHub9 feat/update-routes

Found 1 CRITICAL finding ready to fix:

| Index | Severity | Query | Engine | Location | Status | Files |
|-----|-----|-----|-----|-----|-----|-----|
| 0 | CRITICAL | Stored_XSS | sast | backend/app.py:137 | READY | jinja_filters.py, test_utils.py |

**Summary:** Fix stored XSS vulnerability in role_badge Jinja2 filter

Apply this fix?

>
```

Claude lists the confirmed findings and asks which to apply

# Reviewing each change

Each change is a diff you decide on, and Claude explains itself afterwards.

- Claude Code asks before every edit: **Yes, Yes, and don't ask again this session**, or **No**, and tell Claude **what to do differently**.
- Answering No leaves the file as it was, and you can say what you would rather have.
- When the edits are done, Claude explains the finding and names the tests Remediation Assist generated for it.

```
claude - ProjectHub9

> yes

• Edit(backend/utils/jinja_filters.py)
-     color = role_colors.get(role, 'secondary')
-     return f'<span class="badge badge-{color}">{role}</span>'
+     color = role_colors.get(role, 'secondary')
+     # HTML-encode the role value to prevent Stored XSS when the template uses |safe
+     safe_role = html.escape(str(role) if role is not None else '')
+     return f'<span class="badge badge-{color}">{safe_role}</span>'

Do you want to make this edit to jinja_filters.py?
1. Yes
2. Yes, and don't ask again this session
3. No, and tell Claude what to do differently (esc)
```

Each change is shown as a diff; choose Yes to accept it or No to reject

```
claude - ProjectHub9

• Fixes accepted. Explanation:

Stored XSS in `role_badge` filter (CWE-79)

What: The filter constructs an HTML badge by embedding the `role` value from the database directly into an f-string without HTML encoding. The template applies `|safe` to the output, bypassing auto-escaping. An attacker can store a payload (e.g., `<script>alert(1)</script>`) in the role field to execute arbitrary JavaScript in admin dashboards.

Why: Targets administrators (highest-privilege users). A successful exploit steals admin tokens, exfiltrates credentials, or modifies app data with no further interaction after the payload is stored. Violates OWASP Top 10 (A03:2021).

How: Added `import html` and wrapped the role value with `html.escape()` before embedding in the HTML string. The fix handles `None` input gracefully and neutralizes XSS payloads even when the template applies `|safe`.
```

After the edits, Claude explains what was wrong, why it matters, and how the fix works

# What happens, step by step

The full sequence, from the command to the commit.

- 1 **The developer runs the command.** In Claude Code, `/cx-findings-to-fix:fix`, optionally followed by the project name and branch.
- 2 **Claude runs the tool.** The tool reads the project's git remote and current branch and looks the project up in Checkmarx One.
- 3 **The tool asks when it is unsure.** If the name matches no project, or the branch has no completed scan, Claude shows a numbered list and the developer picks.
- 4 **The tool fetches the fixes.** It lists the Confirmed critical and high findings on the latest completed scan and fetches, at no cost, every fix that already exists. If a finding has no fix yet, Claude asks first: generating one runs Remediation Assist and consumes Checkmarx Credits.
- 5 **Claude shows the findings** as a table and asks which to apply.
- 6 **The tool computes each fix against the current files** without writing anything, producing the complete new content for each changed file.
- 7 **Claude proposes each fix as an edit.** The change is shown as a diff the developer accepts or rejects, file by file.
- 8 **Files that changed since the scan are placed by hand,** so the developer's own edits stay in place.
- 9 **Claude explains each fix** (what, why, how) and names the tests Remediation Assist generated.
- 10 **Claude offers to run those tests** and does so only if the developer says yes. The developer then commits as usual. The plugin never commits or pushes.

## When a file changed since the scan

Claude reads the intended change and the current file, makes the same change where the code now lives, leaves the developer's other edits in place, and says plainly that this fix was placed by hand and deserves a closer look. Claude never overwrites a whole file on its own. Fresh fixes take two to three minutes. Fixes that already exist come back in seconds.

## The whole run, end to end

```
> /cx-findings-to-fix:fix CxRW-Sandbox/ProjectHub9 feat/update-routes
Project CxRW-Sandbox/ProjectHub9, branch feat/update-routes, scan from 2026-08-13.
| Index | Severity | Query          | Location          | Status | Files                                     |
| 0      | CRITICAL | Stored_XSS     | backend/app.py:137 | READY  | jinja_filters.py, test_utils.py |
Apply this fix?
> yes
Edit(backend/utls/jinja_filters.py)
Do you want to make this edit to jinja_filters.py?
1. Yes  2. Yes, and don't ask again this session  3. No, and tell Claude what to do differently
> 1
Fixes accepted. Explanation: What ... Why ... How ...
Checkmarx generated tests in backend/tests/test_utils.py. Run them?
```

# Where things live and what can happen

Which credential is used where, and the boundaries the plugin works inside.

## Where the credential goes and does not go

| Credential                           | Lives                                                          | Used by                                                                         | Never reaches                           |
|--------------------------------------|----------------------------------------------------------------|---------------------------------------------------------------------------------|-----------------------------------------|
| Checkmarx One API key                | shell profile or <code>cx</code> CLI config on the workstation | the <code>ftf</code> tool, which exchanges it for a short-lived token in memory | the assistant's conversation, Anthropic |
| Claude Code session                  | Claude Code's sign-in                                          | Claude Code                                                                     | Checkmarx                               |
| CI/CD credential that triggers scans | the CI/CD system                                               | the pipeline                                                                    | the developer's machine                 |

### What can and cannot happen

- The plugin reads findings and fetches fixes. It cannot change a finding's state, trigger a scan, or touch Checkmarx configuration.
- A fix is generated only after the developer agrees to spend Checkmarx Credits on it. Fixes that already exist are fetched at no cost.
- Every change to code arrives as an edit the developer accepts. The tool writes nothing into the project on its own during the normal flow.
- The plugin never commits or pushes.
- Only Confirmed critical and high findings are considered. Package (SCA) fixes are supported and off by default.
- Nothing runs on a server. Nothing is installed besides the plugin folder.

### Good to know

- It works whether your project folder is a git clone or a plain folder.
- Monorepos: open one service's folder rather than the whole repository and you see only the findings under that folder, with the project's total noted. Ask for "all findings" to see everything.
- Ask Claude to "include package fixes" when you want SCA fixes as well.
- Claude Code's Bash tool has to be allowed. The first run asks you to permit the plugin's command.
- If something is missing (no API key, expired key, feature not enabled on your tenant), Claude tells you exactly what and how to fix it.
- Triage Assist sets the Confirmed and Proposed Not Exploitable states on the platform. Nothing in the plugin changes those verdicts.

# Rolling out to a team

Host the plugin repository somewhere developers can reach, then let them install it in two commands.

## INSTALL

### Each developer runs two commands

The repository can sit in GitHub, Bitbucket, or an internal Git server. Each developer adds it as a marketplace and installs the plugin from it, once:

```
claude plugin marketplace add <repository URL>
claude plugin install cx-findings-to-fix@cx-findings-to-fix-claude
```

## MANAGED SETTINGS

### Pre-approve it centrally

Claude Code's managed settings can pre-approve the marketplace for everyone, so each developer does not have to approve the source themselves.

Per developer, one time: a Checkmarx One API key, Python 3 or Node on the machine, and Claude Code's Bash tool allowed. The first run asks the developer to permit the plugin's command.

## What is in the plugin

Each file has exactly one job.

```
cx-findings-to-fix-claude/
├── .claude-plugin/
│   ├── plugin.json           the manifest: name, version, description
│   └── marketplace.json      makes this repository installable as a marketplace
├── commands/fix.md           the /cx-findings-to-fix:fix command: the protocol
├── agents/findings-to-fix.md a subagent with the same protocol, for delegation
├── skills/fix-confirmed-findings/
│   ├── SKILL.md              the same protocol as a skill, triggered by phrasing
│   ├── ftf.py                 the tool: every deterministic step
│   └── ftf.js                  the identical tool, for machines without Python
├── hooks/                     the session note and the API key warning
└── docs/                       this guide, how it works, architecture page
```

Markdown decides; code does. The command, the subagent, and the skill are one protocol written for three entry points, and they carry the judgment: ask rather than guess, propose rather than write, offer the tests once. The tool carries none: it talks to the Checkmarx One API, computes fixes against your current files, applies nothing on its own, and prints one JSON document. That split is why every run behaves the same way.

Nothing runs on a server. Nothing is installed besides the plugin folder. A GitHub Copilot version of the same plugin exists as a separate repository, `cx-findings-to-fix`.

## Support

Questions and issues: open an issue in the plugin repository.